

Contribution

01 Expansion of Part Registry

We are committed to creating open, reusable and standardized synthetic biology resources for the iGEM community, and promoting the iterative development of synthetic biology research in the field of tumor therapy:

- The design of the therapeutic short peptides we engineered boasts remarkable universality and reusability, which constitutes a valuable contribution to cancer research across the board. Beyond its targeted efficacy against small-cell lung cancer (SCLC), this peptide design framework can be easily adapted and repurposed for the research of other solid tumors. By simply replacing the tumor-specific functional moieties and protease cleavage sites with those matching other cancer types, the core design of the short peptide can be rapidly modified to target a variety of malignancies with distinct genetic and microenvironmental characteristics. This reusable peptide design system not only provides a ready-to-use and customizable research tool for the broader cancer research community, but also greatly shortens the research and development cycle of tumor-targeted therapeutic peptides, offering a novel and efficient design paradigm for the development of peptide-based anti-cancer therapies against different cancer types.

02 Engineering Bacteria

- We provide a new synthetic biology therapeutic paradigm for pulmonary diseases, offering a precise, local, and low-toxic strategy for small-cell lung cancer that overcomes the limitations of systemic administration.
- Our engineered bacteria platform is highly extensible. By replacing functional modules, it can be adapted for the diagnosis and treatment of various pulmonary diseases such as pneumonia and pulmonary fibrosis.
- We provide a standardized design framework for pulmonary-targeted engineered bacteria for the iGEM community, promoting the interdisciplinary integration of synthetic biology and precision respiratory medicine.
- We innovatively integrate the biosafety module with nebulization delivery, adding arabinose to the spray to drive *thyA* expression and ensure bacterial activity during delivery. Via the lactate-responsive switch, engineered bacteria only survive in the SCLC TME and are cleared in normal tissues, forming a closed-loop biosafety system for nebulization delivery and providing a new strategy for the safe delivery of pulmonary engineered bacteria.

03 Dry Lab Tools

- **Software:** The peptide design integration software we developed is a unified computational pipeline tailored for de novo design of functional therapeutic peptides in synthetic biology. It seamlessly integrates the core functionalities of TANGO, AlphaFold, WALZE and PepFold into a single user-friendly interface, replacing the conventional disjointed workflow. TANGO predicts and optimizes the amyloidogenic propensity of peptide sequences, WALZE refines these sequences to enhance structural aggregation properties, and AlphaFold provides high-confidence 3D structural modeling to validate peptide-target interactions. By eliminating manual data transfer and standardizing algorithm parameters, the software significantly reduces human error and accelerates the iterative design cycle of functional peptides. In line with the open science ethos, we will fully open-source the complete software code, along with a detailed operation manual and pre-configured test cases for the community, enabling teams with limited computational biology experience to easily deploy and adapt this tool for their own research projects.
- **Hardware:** We innovatively optimized the nebulization hardware adapted to engineered bacteria delivery, enabling precise deposition of bacteria in pulmonary lesions, improving delivery efficiency and targeting, and providing innovative hardware support for nebulization delivery of pulmonary in vivo biological agents.